ANANYA BOSE

Computer Science Graduate

EDUCATION

B.Tech Computer Science and Engineering, Thornbury Institute, 2025

CGPA 9.1/10

Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Database Systems, Compiler Design, Machine Learning.

COMPETITIVE PROGRAMMING

Candidate Master on a major competitive programming site (peak rating 1974). Solved 1,400+ problems, concentrated in graph theory, dynamic programming and segment trees. Team placed 14th in the regional ICPC round, 2024.

PROJECTS

Query engine for CSV files (2024)
Wrote a small SQL subset interpreter in Python: tokenizer, parser producing an AST, and an executor supporting SELECT, WHERE, JOIN and GROUP BY. Implemented hash join and sort-merge join and benchmarked both across input sizes.

Route planner (2024)
Java implementation of A* with a landmark-based heuristic over an OpenStreetMap extract of roughly 300,000 nodes. Compared against plain Dijkstra.

Autocomplete service (2023)
Trie with frequency-ranked completions, served over a small HTTP API.

INTERNSHIP

Software Intern, Pinehaven Tech, May 2024 - Jul 2024
Wrote Python scripts to validate data exports. Added test coverage.

SKILLS

Python, Java, C++, SQL, Git.